Sri Lanka Institute of Information Technology

Assignment 02 (Individual) – GAMES DEVELOPMENT

SE4031 – Games Development

Title – *2D Interactive Game*

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Weight: ***20%***

IT22582560 - Kahatapitiya K G S M

Summary of the Game Concept

"Bubble Fish 2D" is a 2D interactive underwater survival game. The player controls a brave little yellow fish navigating through a dangerous ocean environment. The core objective is to survive as long as possible while the game automatically scrolls forward. The player must dodge coral reefs (obstacles), shoot down dangerous sea creatures like octopuses and angry fishes (enemies) using bubbles, and collect health packs to stay alive. The game gets progressively harder as the auto-scrolling speed increases over time.

Implementation of Core Features

- **Health System:** The player starts with a maximum health value of 100. Health is displayed dynamically on the UI. Colliding with an obstacle (Coral Reef) decreases health by 10 , and colliding with an enemy decreases it by 20. When the player hits an obstacle, they pass through it while taking damage. The game effectively stops player movement when the health reaches 0 , and the health resets to maximum upon restarting the game.
- **Score System:** The score is displayed as a numeric value on the UI. The score increments when the player successfully shoots and destroys an enemy or survives over time. It resets to 0 when the game is restarted.
- **Health Packs:** Health packs are collidable objects visually distinct from enemies. Upon collision with the player, they increase the health value up to the max value of 100 and are immediately destroyed. If missed, they are automatically destroyed after passing the camera bounds to optimize performance.
- **Projectile Attacks:** The player can shoot bubble projectiles by clicking the left mouse button. The projectile spawns at the player's current position and moves forward. It is destroyed upon colliding with an enemy or after a short duration.
- **Enemies:** Enemies are collidable objects that spawn randomly ahead of the player. They decrease the player's health upon collision. They are destroyed if hit by a player's projectile , which rewards the player with an increased score.
- **Speed Increment:** To increase the difficulty smoothly, the auto-scrolling speed of the game steadily increases over time using Time.deltaTime.

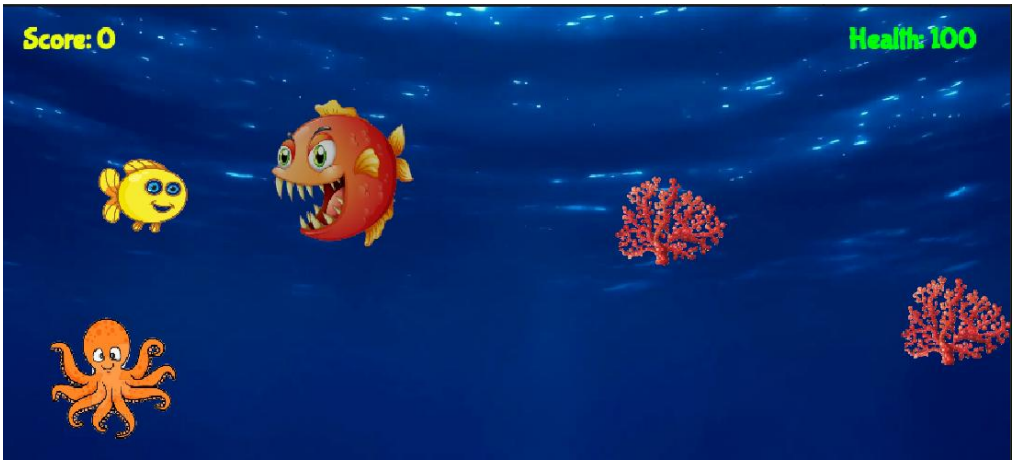
Control Guide

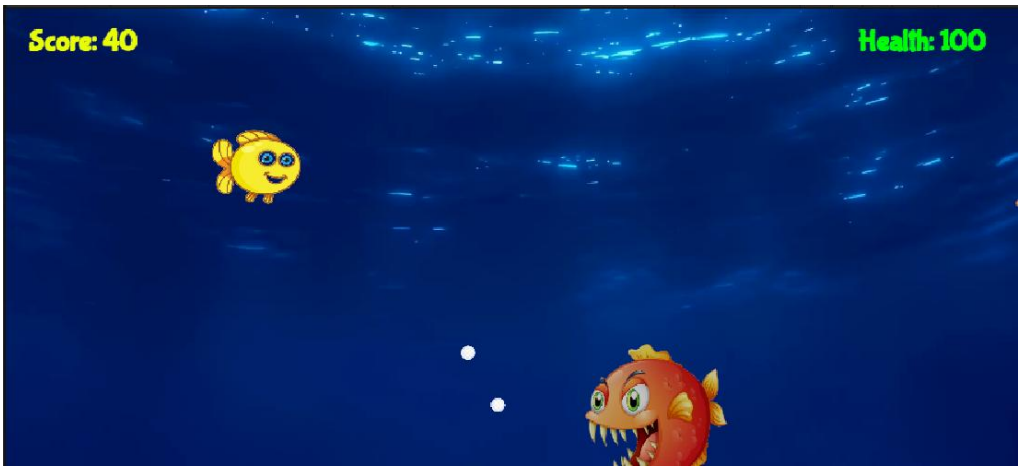
- **W / Up Arrow:** Move Up
- **S / Down Arrow:** Move Down
- **A / Left Arrow:** Move Backward
- **D / Right Arrow:** Move Forward
- **Left Mouse Click:** Shoot Bubbles (Projectile)

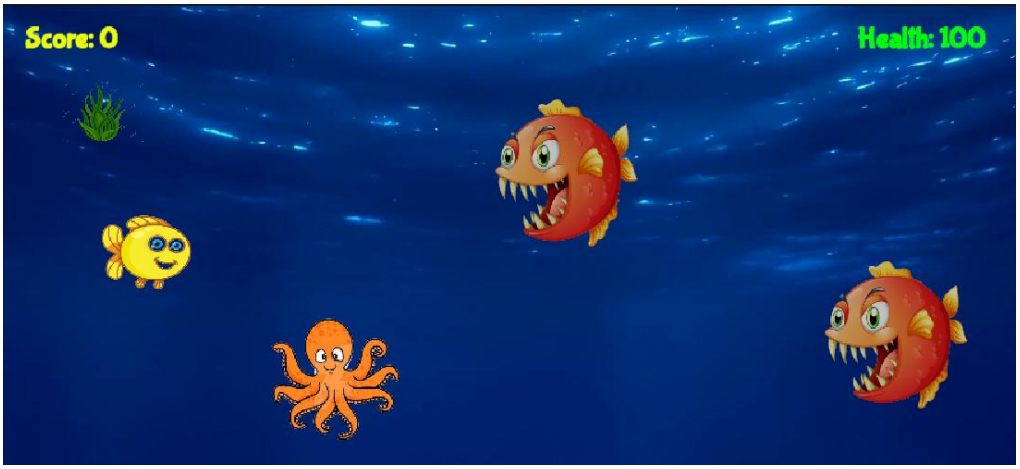
Creative Feature

Dynamic UI Health Indicator & Custom Audio System As a creative modification, the game's Health UI constantly updates its color based on the player's current health status (Green for High Health, Yellow for Medium, and Red for Critical). Furthermore, a custom audio manager was implemented with individual volume controls for different events, featuring unique sound effects for taking damage from obstacles vs. enemies, collecting health, and shooting, layered over a continuous underwater background video and track.

Screenshots of Gameplay



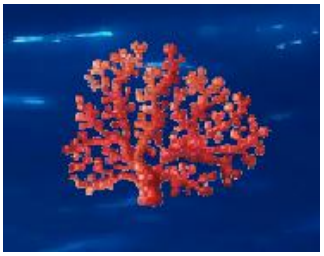




Player

Enemy

Obstacle



HealthPack

Bubbles

Credits and External Assets

- **Game Engine:** Unity (2022.3)
- **Visual Assets (Sprites & Background):** Downloaded from Freepik / Adobe Stock (Free License).
- **Audio Assets (Music & SFX):** Downloaded from Pixabay (Royalty-Free Sound Effects).

Gameplay Demo Video Link

https://drive.google.com/file/d/1fp5KFCUOVrqdHi_o21Td7f_ZVVk8mSBp/view?usp=sharing